

HCMC UNIVERSITY OF TECHNOLOGY – VNU-HCM
FACULTY OF CHEMICAL ENGINEERING
DEPARTMENT OF BIOTECHNOLOGY

FERMENTATION EXPERIMENTS

For the internal circulation only

WEEK 1: PREPARATION OF FERMENTATION & MICROORGANISMS

Bacteria strain: *Edwardsiella tarda* E24.1

Phage strain: *E. tarda* phage DT115P1

Preparation

- TSB medium
 - o 3% (w/v) TSB
- 1.5% agar TSB medium
 - o 3% (w/v) TSB
 - o 1.5% (w/v) agar
- 0.4% agar TSB medium
 - o 3% (w/v) TSB
 - o 0.4% (w/v) agar
- PBS buffer + 20% glycerol
 - o 0.8% (w/v) NaCl
 - o 0.02% (w/v) KCl
 - o 0.288% (w/v) Na₂HPO₄·12H₂O
 - o 0.02% (w/v) KH₂PO₄
 - o 20% (v/v) glycerol
- SM buffer
 - o 0.58% (w/v) NaCl
 - o 0.2% (w/v) MgSO₄·7H₂O
 - o 5% (v/v) Tris.HCl 1M (pH=7.5)
 - o 2% (v/v) gelatin
- Petri plate (Ø90 mm)
- 100 mL, 250 mL Erlenmeyer flask
- 50 mL falcon
- 1.5 mL, 2.0 mL Eppendorf
- 5 mL sterile syringe
- 0.45 µm sterile syringe filter

Bacteria recover

- From *E. tarda* E24.1 freezing stock, use sterile loop for culturing on 1.5% agar TSB plate.
- Incubate overnight at 28 °C (room temp).
- Pick individual bacterial colony to 15 mL TSB medium in falcon.
- Incubate with agitation overnight at 28 °C (room temp) to obtain maximum cell growth.

Report:

- Summary the lesson prior to attending the class
- Take note the changes during the and results/discussion after the class

WEEK 2 + 3: BACTERIA STORAGE AND PHAGE RECOVER

Bacteria storage

- *E. tarda* E24.1 culture was incubated overnight.
- In 1.5 mL Eppendorf, 500 μL culture was mixed with an equal volume of PBS buffer (+20% glycerol), the final proportions of bacterial stock should be 10% (v/v) diluted glycerol.
- Mix gently to yield a homogenous solution.
- Place the tubes in the freezer at $-70\text{ }^{\circ}\text{C}$.

Phage recover

Step 1: Drop assay:

- 5ml TSB 0.4% agar is boiled and then maintained at $43\text{ }^{\circ}\text{C}$.
- Mix 200 μL *E. tarda* E24.1 culture in 5ml TSB 0.4%, then pour into the plate, keep for about 2 min
- Drop 2 μL of the phage filtrate onto the plate, keep for $\sim 2\text{ mins}$
- Incubate 18–20 hour at $28\text{--}30\text{ }^{\circ}\text{C}$ (room temp) $\rightarrow$ transparent zone will appear on the plate.

Step 2

- Pick a part of transparent zone by micropipette tip and transfer into 10 μL SM buffer, vortex in 2 mins. Centrifuge at 12,000rpm, $4\text{ }^{\circ}\text{C}$ for 5 mins. Transfer 5 μL supernatant into 195 μL SM buffer, and dilute to 10^{-1} , 10^{-2} , 10^{-3} . Do plaque assay: Mix 100 μL of each diluted phage lysate and 200 μL *E. tarda* E24.1 culture in 5ml TSB 0.4% agar (that was boiled and maintained at $43\text{ }^{\circ}\text{C}$), then pour into plate $\rightarrow$ incubation overnight at room temp.

Step 3 – Harvesting: After nearly-confluent lysis (choose the plate similar as the below figure)

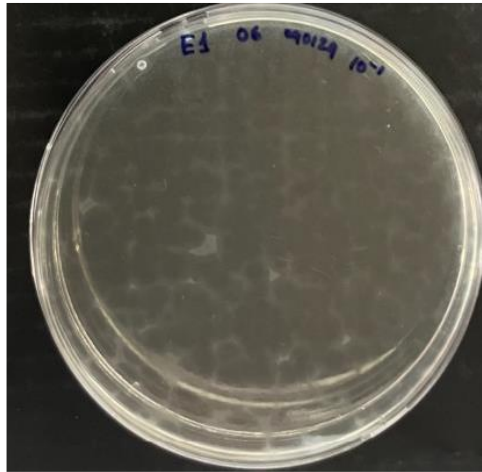


Fig.1. High titer-phage plate

- Place the plate face-up and load the plate with 4mL SM buffer (for ϕ 90mm plate), incubate the plate in 4 °C overnight.
- Collect the supernatant and centrifuge at 12,000rpm, 4 °C for 5 min. The phage-containing supernatant is filtrated by 0.2 um membrane filter to obtain phage lysate.
- Phage lysate can be stored at 4 °C or stored at -70 °C (by mixing with SM buffer 20% glycerol).

Phage titration by plaque assay

- Serial dilute phage lysate with SM buffer. Mix 100 μ L of phage lysate and 900 μ L of SM buffer. Briefly vortex the tube. Repeat for each dilution.
- Mix 100 μ L phage lysate and 200 μ L overnight *E. tarda* E24.1 culture in 5 ml 0.4% agar TSB, then pour into 1.5% agar TSB plate.
- After 6 – 8h incubation at 28 °C (room temp), plaques became visibly distinguishable.
- Count the number of plaques on the plate to calculate the phage titer.

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WEEK 4: PHAGE PROPAGATION

Phage propagation via liquid cultivation

- Grow *E. tarda* E24.1 bacterial host in TSB medium overnight at 28 °C (room temp).
- Prepare 50 ml of TSB medium in Erlenmeyer flask 250 mL.
- Add 2% (v/v) of overnight *E. tarda* E24.1 culture.
- Incubate with agitation for 1h at 28 °C to reach $OD_{595nm} = 0.1$ ($\sim 2 \cdot 10^8$ CFU/ml).
- Add 100 μ L of phage lysate (MOI= 0.1).
- Incubate at 28 °C (room temp) with agitation for 6h or until lysate clears.
- Collect phage lysate and store at 4 °C until clean up.
- Aliquot phage lysate into 2.0 mL Eppendorf and centrifuge at $12,000 \times g$ for 5 min.
- Carefully collect supernatant using a syringe.
- Filter-sterilize the phage supernatant using a 0.45 μ m filter to yield a bacterial cell-free phage lysate.
- Titer the lysate with plaque assay to ensure initial high titer.

Phage propagation on agar plate

- Transfer 5 μ L supernatant into 195 μ L SM buffer, and dilute to 10^{-1} , 10^{-2} , 10^{-3} . Do plaque assay: Mix 100 μ L of each diluted phage lysate and 200 μ L *E. tarda* E24.1 culture in 5ml TSB 0.4% agar (that was boiled and maintained at 43 °C), then pour into plate.
- Harvesting: After nearly-confluent lysis (choose the plate similar as the below figure)

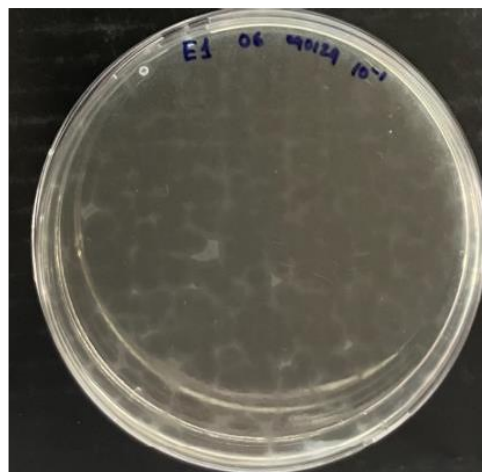


Fig.1. High titer-phage plate

- Place the plate face-up and load the plate with 4mL SM buffer (for ϕ 90mm plate), incubate the plate in 4°C overnight.
- Collect the supernatant and centrifuge at 12,000rpm, 4°C for 5 min. The phage-containing supernatant is filtrated by 0.45 μ m membrane filter to obtain phage lysate.

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WEEK 5: PRODUCT CHECK AND STORAGE

Phage titration by plaque assay method

- Serial dilute phage lysate with SM buffer. Mix 100 μ L of phage lysate and 900 μ L of SM buffer. Briefly vortex the tube. Repeat for each dilution.
- Mix 100 μ L phage lysate and 200 μ L overnight *E. tarda* E24.1 culture in 5 ml 0.4% agar TSB, then pour into 1.5% agar TSB plate.
- After 6 – 8h incubation at 28 °C, plaques became visibly distinguishable.
- Count the number of plaques on the plate to calculate the phage titer.

Phage storage

- Phage lysate can be stored at 4 °C or stored at -70 °C (by mixing with SM buffer 20% glycerol).

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